

# Ismael Afolabi ASSANI

Senior Quantitative Analyst – Model Validation, Risk Analytics & Financial Modelling

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## Profile

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Quantitative analyst with a Ph.D. in Statistics, specialized in financial risk modelling, model validation, derivative pricing, econometrics, forecasting, and quantitative risk analytics. Experienced in assessing model assumptions, testing model performance, developing benchmark models, conducting sensitivity analysis, stress testing, backtesting, and documenting model limitations. Doctoral research in mean-variance hedging, basis risk, pricing kernels, GARCH models, stochastic interest rates, and multi-asset risk models, with implementation in Python, R, Matlab, and SQL.

## Core Skills for Quantitative Model Validation

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- **Model Validation & Governance:** independent model review, conceptual soundness assessment, benchmark models, sensitivity analysis, backtesting, stress testing, performance monitoring, model limitations, controls assessment, and validation reporting.
- **Financial Risk Models:** market risk, credit risk, liquidity risk, interest rate risk, VaR, Expected Shortfall, tail risk, crisis scenarios, funding gaps, maturity ladders, and portfolio exposures.
- **Pricing, Econometrics & Forecasting:** derivative pricing, risk-neutral valuation, pricing kernels, ARIMA, multivariate GARCH, HMM/IOHMM, Kalman filters, regression, maximum likelihood estimation, Monte Carlo simulation, and model comparison.
- **Quantitative Methods:** statistics, probability, linear algebra, optimization, PCA, covariance modelling, numerical methods, simulation, empirical testing, and reproducible quantitative research.
- **Tools:** Python, pandas, NumPy, SciPy, statsmodels, scikit-learn, Matlab, R, SQL, SAS, Stata, Excel/VBA, Power BI, Git, and  $\text{\LaTeX}$ .

## Relevant Experience

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### Principal Analyst – Treasury Risk

Nov. 2025 – Dec. 2025

*Business Development Bank of Canada (BDC) – Montreal, Canada*

- Developed and maintained quantitative models for interest rate risk, liquidity risk, funding risk, and portfolio-level exposures.
- Built Python pipelines to process financial data, generate scenarios, and compute risk metrics: DV01, duration, convexity, cash-flow sensitivities, funding gaps, and maturity ladders.
- Prepared risk reports for senior management and ALCO, summarizing assumptions, model limitations, quantitative results, and business implications.

### Intern – Energy Forecasting Analyst

Feb. 2021 – Jun. 2021

*Plant-E Corp – Montreal, Canada*

- Developed R/Python pipelines for hourly and multi-horizon electricity price and volatility forecasting using GARCH, LSTM, Prophet, HMM, and IOHMM models.
- Performed data cleaning, feature engineering, out-of-sample validation, and performance monitoring using MAE, RMSE, MAPE, and model comparisons.
- Conducted regime analysis and scenario design to support decision-making in a volatile energy market.

### Statistician Economist

Oct. 2017 – Jul. 2018

*BCEAO – Central Bank of West African States – Dakar, Senegal*

- Built business-cycle dashboards and monitored macro-financial indicators using R, Stata, and Excel/VBA.
- Processed surveys, performed quality control, integrated datasets, and drafted economic and financial analysis notes.

- Lecturer in linear regression, concepts and methods in statistics, and applied quantitative methods.
- Teaching assistant in mathematical finance, derivatives and risk management, forecasting methods, statistics, and actuarial science / finance courses.

## Doctoral Research and Relevance to Model Validation

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**Doctoral research:** development and validation of discrete-time financial models for hedging and basis-risk management, including semi-explicit hedge ratios, GARCH dynamics, stochastic interest rates, risk-neutral pricing kernels, and multi-instrument hedging portfolios. The work combines theoretical model development, mathematical validation, numerical implementation, empirical testing, backtesting, and risk analytics.

- Assessed model assumptions and limitations in incomplete markets: basis risk, imperfect correlation, stochastic volatility, leverage effects, and stochastic interest rates.
- Built benchmark models and alternative strategies to compare risk-minimizing, variance-optimal, delta-based, and naive hedging approaches.
- Implemented numerical pricing and hedging methods: optimization, inverse transforms, quadrature, Monte Carlo comparisons, and out-of-sample evaluation.

## Publications and Working Papers

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- Assani, I.; Cissé, M.; Konte, M.; Touré, M. (2019). [Contribution to the Valuation of BRVM Assets: A Conditional CAPM Approach](#). *Journal of Risk and Financial Management*, 12(1), 27. Conditional CAPM, time-varying beta, Kalman filter, Markov-switching models, SMB factor, and RMSE model comparison.
- Assani, I.; Augustyniak, M.; Badescu, A.; Bégin, J.-F.; Stentoft, L. (2025). [Discrete-time hedging, basis risk, and covariance-dependent pricing kernels](#). Working paper / submitted. Semi-closed-form risk-minimizing hedge ratios under basis risk with bivariate GARCH dynamics; empirical analysis with S&P 500, futures, and VIX data.
- Assani, I.; Augustyniak, M.; Godin, F. (2026). [Discrete-time variance-optimal hedging with basis risk for multi-asset hedging portfolios](#). Working paper / submitted. Semi-explicit variance-optimal hedging formulas for multi-instrument portfolios under basis risk, fast quadrature-based computation, and hedging-effectiveness analysis.

## Education

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**Ph.D. in Statistics**, Université de Montréal  
Montreal, Canada

2018 – 2025

- Thesis: [Hedging of derivative products in the presence of basis risk](#). Accepted in April 2025.
- Areas: incomplete markets, local and global quadratic hedging, GARCH models, risk-neutral measures, stochastic interest rates, and mean-variance hedging.

**Engineer Statistician Economist**, ENSAE Dakar  
Dakar, Senegal – Master's equivalent

2014 – 2017

**Bachelor's in Statistics and Applied Economics**, ENEAM  
Cotonou, Benin

2010 – 2013

## Selected Additional Experience

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**CRES, Dakar – Survey Controller (2018):** field supervision, protocol monitoring, quality control, and data validation in R/Stata. **CREFDES, Dakar – Research Assistant (2017–2018):** UEMOA output-gap estimation using state-space models and Kalman-filter methods. **DPEE, Dakar – Research Assistant (2016):** fixed-effects panel analysis, policy evaluation, and economic reporting. **DGAE, Cotonou – Research Assistant (2013):** gravity models, customs integration, macroeconomic analysis, and documentation.

## Awards and Conferences

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FRQNT doctoral scholarship (2021–2023); FONCER Fin-ML / IVADO (2020–2021); Université de Montréal Department of Mathematics and Statistics scholarships (2018–2021). Presentations: SSC 2023, ISM 2023, Optimization Days 2022, FinML 2021.

## References

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**Prof. Maciej Augustyniak** – Associate Professor, Department of Mathematics and Statistics, Université de Montréal  
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**Francis Andrianarison, Ph.D.** – Senior Economist, United Nations Development Programme (UNDP)  
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